

Physical and Chemical Properties of Soil Near Cracks in Irrigated Vertisols Sown with Cotton-Wheat Rotations

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*Properties of soil adjacent to cracks and in bulk soil were compared in furrow irrigated Vertisols sown to intensively tilled cotton (*Gossypium hirsutum* L.) followed by (fb.) minimum-tilled wheat (*Triticum aestivum* L.) in a 2-year rotation; minimum-tilled cotton fb. and minimum-tilled wheat in a 2-year rotation; and a perennial pasture in NW New South Wales, Australia. A backhoe pit was dug at right-angles to cracks, and a 50 mm thick layer of soil sampled from the exposed crack walls and from adjacent bulk soil in 0.3 m depth increments to a depth of 0.9 m. Soil properties evaluated were: aggregate stability (dispersion index), soil resilience to structural destruction (as geometric mean diameter of aggregates formed after puddling and drying of soil), electrical conductivity ($EC_{1:5}$), exchangeable cations, $EC_{1:5}$ /exchangeable Na ratio, CEC, $CaCO_3$ equivalent, pH, and organic C. Intensive tillage resulted in soil adjacent to cracks having lower exchangeable Ca, CEC, and $CaCO_3$ equivalent, and higher organic C and plastic limit than bulk soil away from cracks. More stable aggregates also occurred with intensive tillage. With minimum tillage, only plastic limit and organic C in the 0.6–0.9 m depth of soil adjacent to cracks were higher, and $CaCO_3$ in all depths lower compared with bulk soil. In the untilled pasture, $EC_{1:5}$ in the 0.0–0.3 m depth and $EC_{1:5}$ /exchangeable Na ratio in the 0.0–0.6 m depth were lower in soil adjacent to cracks than in bulk soil. We suggest that the differences between soil adjacent to cracks and bulk soil under intensive tillage, and similarities between soil adjacent to cracks and bulk soil under pasture and minimum tillage were due to the presence of stable biopores in the latter group. Biopores can function as an alternative pathway to soil cracks for root growth, preferential flow, and associated nutrient leaching. However, when the biopore networks are disrupted by tillage and cracks become the dominant pathway for preferential flow, nutrient leaching, and root growth, significant differences in physical and chemical properties can occur between soil adjacent to cracks and bulk soil.*

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Vertisols, which are dominated by smectitic clay minerals, swell on wetting and form deep cracks on drying (McGarity, Hoult, and So 1984; Probert et al. 1987). Soil crack formation is facilitated by rapidity and intensity of profile drying (Bauder and Brock 1992) and by increasing the number of wetting/drying cycles (Pillai and McGarry 1999). Cracking can improve soil structure, facilitate water infiltration and drainage, and help to distribute roots, soil organic matter, and nutrients in the soil profile (Dexter 1986a, 1991; McGarity, Hoult, and So 1984; Probert et al. 1987; Skjemstad, Janik, and Taylor 1998; Van Noordwijk et al. 1993). This concept has been applied to managing soil quality in Vertisols by sowing crop sequences which maximize soil drying and cracking, such as wheat (*Triticum aestivum* L.) or safflower (*Carthamus tinctorius* L.) after irrigated cotton (*Gossypium hirsutum* L.) (Cooper 1999; McGarity, Hoult, and So 1984). Cracking also helps to leach nutrients, fertilizers, herbicides, and pesticides (Baer et al. 1995; Haria et al. 1994). Given their central role in these processes, physical and chemical properties of soil lining the cracks should differ significantly from those of nearby soil away from cracks. Research on soil dynamics of Vertisols suggest that this occurs with respect to soil shrinkage characteristic (McGarity, Hoult, and So 1984; Probert et al. 1987).

Are there then other soil properties, particularly more easily measured properties, which also differ between soil adjacent to cracks and bulk soil? Published data on physical and chemical properties of soil adjacent to cracks (i.e., from crack walls) are sparse. Sachan and Sharma (1980) observed that with respect to soil organic matter fractions, soil near cracks tended to have higher proportions of humic acids relative to fulvic acids, and the reverse was true as distance from cracks increased. Other studies have reported that soil organic matter, roots, soil bacteria and protozoan populations, and soil processes such as mineralization occurred at higher rates closer to soil cracks than further away (Van Noordwijk et al. 1993). Both groups of workers speculated that these responses might be caused by better aeration of soil proximate to soil cracks. Crop residues and organic matter falling in cracks in Vertisols may also result in organic carbon values of soil adjacent to cracks being higher than that in bulk soil (Skjemstad, Janik, and Taylor 1998). The objective of this study, therefore, was to quantify and compare physical and chemical properties of bulk soil with that adjacent to cracks formed under irrigated cotton-wheat rotations in eastern Australian Vertisols.

Materials and Methods

One experimental site was located on-farm at "Glenarvon," Wee Waa (149°27'E, 30°13'S) and two others at the Australian Cotton Research Institute (ACRI) in Narrabri, New South Wales, Australia (149°47'E, 30°20'S) in the Namoi valley of New South Wales. Two of the sites ("Glenarvon," 400 m × 24 m; and Field C1 at ACRI, 200 m × 16 m) had been sown with a furrow-irrigated cotton-wheat 2-year rotation since 1993, whereas the third site at ACRI was a 25-year-old permanent pasture, 80 m × 10 m, dominated by prairie grass (*Bromus catharticus* L.), soursob (*Oxalis pes-caprae* L.) and burr medic (*Medicago polymorpha* L.). Land preparation at "Glenarvon" consisted of reduced tillage, where wheat was sown after minimum tillage (only ridge cultivation to a depth of 0.1 m with an aer-way cultivator, a secondary tillage tool consisting of a single bar of rotating tines) and cotton was sown after intensive tillage (disc- and chisel ploughing every year followed by ridging). In Field C1 at ACRI, wheat was sown with no-tillage and cotton after minimum tillage (only ridge cultivation with an aer-way cultivator, followed by disc-hilling and sowing into the reformed ridges). Field C1 had been minimum-tilled since 1985. In both cropped

sites, wheat was sown in early June and harvested in December, whereas cotton was sown in early October and harvested in early May. The experimental sites have a semiarid climate and experience four distinct seasons with a mild winter and a hot summer. The hottest month is January (mean daily maxima of 34°C and minima of 19°C), whereas July is the coldest (mean daily maxima 18°C and minima of 3°C). Mean annual rainfall is 616 mm with a major peak occurring between October and February and a minor peak between June and August, although rainfall distribution and amounts are highly variable. Mean annual potential evaporation (Class A pan) is 1939 mm. The soils are deep, uniform grey clays, and were classified as fine, thermic, montmorillonitic, Typic Haplusterts (Soil Survey Staff 1996). Clay mineralogy consists predominantly of montmorillonite with subdominant fractions of kaolinite and illite. Particle size distribution in the surface 0.9 m at "Glenarvon" was 550 g kg⁻¹ clay (< 2 µm), 180 g kg⁻¹ silt (2–20 µm) and 270 g kg⁻¹ sand (20–2000 µm); in Field C1 it was 560 g kg⁻¹ clay, 150 g kg⁻¹ silt and 290 g kg⁻¹ sand; and in the pasture it was 600 g kg⁻¹ clay, 210 g kg⁻¹ silt and 190 g kg⁻¹ sand.

Eight sampling plots (50 m × 24 m) were established at "Glenarvon" and four in Field C1 (50 m × 16 m). Sampling plots were not established in the pasture site at ACRI. A single crack was selected at random from within plant rows in each sampling plot at "Glenarvon" and Field C1, whereas four cracks were selected at random from the pasture. The first two sites were sampled in late November 1997 and the last in early February 1999. A backhoe pit, approximately 2 m wide, was dug to a depth of 2 m at right angles to each crack. Crack depth and width in the surface and subsurface (0.15–0.90 m) were measured, and soil was sampled from the pit face. A 50 mm thick layer of soil was sampled from the exposed crack walls and from adjacent bulk soil, 0.3–0.5 m from the crack, at depth intervals of 0–0.3 m, 0.3–0.6 m, and 0.6–0.9 m from the soil surface. The samples were then transported to the laboratory for further analysis.

The soil was air-dried and passed through a sieve with 2-mm diameter apertures, and pH (in 0.01M CaCl₂), electrical conductivity of a 1:5 soil:water suspension, CaCO₃ equivalent by titration, and cation exchange capacity as the sum of exchangeable Ca²⁺, Mg²⁺, K⁺, and Na⁺ determined after washing with aqueous alcohol and aqueous glycerol to remove soluble salts and after extraction with alcoholic 1M NH₄Cl at a pH of 8.5 (Rayment and Higginson 1992). Plastic limit, the water content at which a soil changes from a plastic state to a friable state, was determined with a drop-cone penetrometer (30° cone with a mass of 80 grams dropped from a height of 50 mm) on samples prewetted slowly to different water content and equilibrated over a 7-day period at 30°C without remolding in sealed plastic containers (30 cm³) (Weaver and Hulugalle 2000). Soil resilience, a measure of the self-mulching ability of the soil, was determined by puddling 25 g of air-dried soil (< 2 mm) until it became a smooth paste and oven-drying at 40°C for 72 h. The size distribution of the aggregates formed (determined by dry sieving on a mechanical shaker at 1440 vibrations per minute for five minutes) was expressed as the geometric mean diameter of the soil aggregates (Hulugalle, Larsen, and Henggeler 1996). Total soil organic carbon was determined by the wet oxidation method of Walkley and Black on soil which had been passed through a sieve with aperture diameters of 0.5 mm (Rayment and Higginson 1992). All data were statistically analyzed with the MSTAT-C software package (Michigan State University 1993) using a *t*-test where comparisons were made between each pair of crack wall soil and bulk soil. Quantitative comparisons were not made between sites.

Dispersion (after immersion in bore water of EC = 0.4 dS m⁻¹, SAR = 1.7) was determined with a sediment density-specific gravity meter (Mettler-Toledo 110M)¹ on soil aggregates of 1–4 mm diameter previously wetted in a humidifier to soil

¹ Trade names are given for the benefit of the reader and do not mean endorsement of the products by the authors.

water contents ranging from 300 g kg⁻¹ to air-dry (40 g kg⁻¹) (Entwistle Hulugalle, and Finlay 1997). Dispersion index (in g kg⁻¹) was expressed as:

$$\text{Dispersion Index} = \frac{\text{Mass of soil particles} < 20 \mu\text{m released into the suspension due to immersion in water}}{\text{Mass of soil particles} < 20 \mu\text{m released into suspension after complete dispersion of sample}} \times 1000$$

Dispersion index was plotted against soil water content, and the data fitted to equations of the form $DI = a + b \ln \theta$, where DI = dispersion index (g kg⁻¹), a = intercept, b = slope, and θ = soil water content (g kg⁻¹). The effects of experimental treatments on dispersion was evaluated by comparing best-fit dispersion curves with analysis of variance for individual treatments (soil adjacent to cracks vs. bulk soil) for each depth and site.

Results and Discussion

Mean depth ($\pm$ standard deviation) of cracks at "Glenarvon," Field C1, and the degraded pasture at ACRI were 0.86 ± 0.08 m, 0.74 ± 0.05 m, and 1.05 ± 0.12 m, respectively. In all sites, crack width ranged from 20–100 mm in the surface and 5–150 mm in the subsurface (0.15–0.90 m). Surface crack width under degraded pasture was narrowest (27.5 ± 17.1 mm).

At "Glenarvon," CEC in the 0.3–0.9 m depth and CaCO₃ equivalent in all depth were significantly lower, and soil organic C in the 0.3–0.9 m depth and plastic limit in all depths were higher in soil adjacent to cracks in comparison with bulk soil (Table 1). Furthermore, mean exchangeable Ca²⁺ values in the 0–0.9 m depth of soil adjacent to cracks were lower ($P < 0.05$) than those in the corresponding depth of the bulk soil. Mean values of exchangeable Ca²⁺ in soil adjacent to cracks and in bulk soil were 16.9 and 17.3 cmol_c kg⁻¹, respectively. Dispersion in all depths at "Glenarvon" was greater in bulk soil than in soil adjacent to cracks (Table 2). Furthermore in the 0.3–0.6 m and 0.6–0.9 m depths only for bulk soil did dispersion index vary as a function of soil water content (Table 2). The curves observed in the soil adjacent to cracks from these depths were, in a statistical sense, horizontal lines and suggest that soil water content had a negligible effect on aggregate stability. This contradicts published data (Entwistle, Hulugalle, and Finlay 1997; Hulugalle, Entwistle, and Lobry de Bruyn 1997), and may be caused by incomplete wetting of the aggregates due to the hydrophobic nature of some soil organic matter (Bechmann 1996) which was higher in the soil adjacent to cracks (Table 1). Microflora associated with soil organic matter are also thought to confer hydrophobic behavior to soil (Chan 1992).

In Field C1 at ACRI, only plastic limit and soil organic C in the 0.6–0.9 m depth and CaCO₃ in all depths differed significantly between soil adjacent to cracks and bulk soil (Table 3). Plastic limit and organic C were significantly higher and CaCO₃ lower in soil adjacent to cracks than in bulk soil. This response pattern was similar to that shown at "Glenarvon" by these two indices. Significant differences did not occur in Field C1 between soil adjacent to cracks and bulk soil with respect to other soil physical and chemical properties measured. Dispersion curves did not differ significantly between soil adjacent to cracks and bulk soil in any depth. Dispersion curves fitted to data from both soil adjacent to cracks and bulk soil from the 0.0–0.3 m, 0.3–0.6 m, and 0.6–0.9 m depths were as follows (DI = dispersion index in g kg⁻¹; θ = soil water content in g kg⁻¹; s.e.(b) = standard error of the slope; r = correlation coefficient; n = number of observations; *** = significant at the 0.001 probability level; ns = nonsignificant):

TABLE 1 Effect of cracks on soil physical and chemical properties at "Glenarvon," November 1997

Soil property	Location	0.0–0.3 m	0.3–0.6 m	0.6–0.9 m
GMD, mm ^a	Soil adjacent to cracks	5.6	5.9	6.4
	Bulk soil	5.6	6.6	6.5
	<i>P</i> <	ns ^b	ns	ns
pH (0.01M CaCl ₂)	Soil adjacent to cracks	7.0	7.2	7.3
	Bulk soil	7.0	7.2	7.4
	<i>P</i> <	ns	ns	ns
EC _{1:5} ^c , dS m ⁻¹	Soil adjacent to cracks	0.06	0.07	0.11
	Bulk soil	0.06	0.07	0.12
	<i>P</i> <	ns	ns	ns
Plastic limit, g kg ⁻¹	Soil adjacent to cracks	220	216	213
	Bulk soil	218	207	204
	<i>P</i> <	0.05	0.05	0.05
Soil organic C, g kg ⁻¹	Soil adjacent to cracks	9.8	6.1	5.2
	Bulk soil	9.0	5.4	4.4
	<i>P</i> <	0.05	0.01	0.01
CaCO ₃ equivalent ^d g kg ⁻¹	Soil adjacent to cracks	37	34	44
	Bulk soil	43	45	53
	<i>P</i> <	0.05	0.05	0.05
EC _{1:5} /Exch. Na ⁺	Soil adjacent to cracks	0.12	0.07	0.09
	Bulk soil	0.13	0.07	0.09
	<i>P</i> <	ns	ns	ns
CEC ^e , cmol _c kg ⁻¹	Soil adjacent to cracks	27.5	29.6	30.0
	Bulk soil	28.0	30.3	31.3
	<i>P</i> <	ns	0.05	0.05

^aGeometric mean diameter of aggregates formed after puddling and drying an air-dried sample.

^bNonsignificant.

^cElectrical conductivity of a 1:5 soil:water suspension.

^dCalcium and magnesium carbonates. Calcium carbonate accounts for > 90% of the carbonates in these soils.

^eCation exchange capacity.

0.0–0.3 m: DI = 227.2 – 58.11(ln θ), s.e.(b) = 13.3, *r* = – 0.68***, *n* = 24;

0.3–0.6 m: DI = 6.9 + 9.2(ln θ), s.e.(b) = 6.0, *r* = 0.31 ns, *n* = 24;

0.6–0.9 m: DI = 383.8 – 128.8(ln θ), s.e.(b) = 8.2, *r* = – 0.96***, *n* = 24.

In the perennial pasture at ACRI, EC_{1:5} in the 0.0–0.3 m depth and EC_{1:5}/exchangeable Na⁺ ratio in the 0.0–0.6 m depth were lower in soil adjacent to cracks than in bulk soil (Table 4). Dispersion curves did not differ significantly between soil adjacent to cracks and bulk soil in any depth. Dispersion curves fitted to data from both soil adjacent to cracks and bulk soil from the 0.0–0.3 m, 0.3–0.6 m, and 0.6–0.9 m depths were as follows (DI = dispersion index in g kg⁻¹; θ = soil water content in g kg⁻¹; s.e.(b) = standard error of the slope; *r* = correlation coefficient; *n* = number of observations; ** = significant at the 0.01 probability level; ns = nonsignificant):

0.0–0.3 m: DI = 228.4 – 18.8(ln θ), s.e.(b) = 29.7, *r* = – 0.11 ns, *n* = 30;

0.3–0.6 m: DI = 241.0 – 56.5(ln θ), s.e.(b) = 17.5, *r* = – 0.51**, *n* = 30;

0.6–0.9 m: DI = 246.9 – 60.7(ln θ), s.e.(b) = 15.7, *r* = – 0.58**, *n* = 30.

TABLE 2 Dispersion curves in the 0.0–0.3 m, 0.3–0.6 m, and 0.6–0.9 m depths of soil adjacent to cracks and in bulk soil at “Glenarvon”

Depth (m)	Soil	Dispersion curve ^a	Analysis of variance
0.0–0.3	Adjacent to crack	DI = 54.8 – 10.8(ln θ), s.e.(b) = 3.7, r = – 0.53**, n = 24	Total regression: $P < 0.05$; Intercepts: $P < 0.01$; Slopes: $P = \text{ns}$
	Bulk	DI = 68.1 – 12.3(ln θ), s.e.(b) = 2.7, r = – 0.69***, n = 24	
	Adjacent to crack	DI = 46.2 + 1.1(ln θ), s.e.(b) = 2.4, r = 0.10 ns, n = 24	Total regression: $P < 0.01$; Intercepts: $P = \text{ns}$; Slopes: $P < 0.01$
0.3–0.6	Adjacent to crack	DI = 68.4 – 9.0(ln θ), s.e.(b) = 2.0, r = – 0.69***, n = 24	
	Bulk	DI = 40.6 + 1.7(ln θ), s.e.(b) = 6.0, r = 0.18 ns, n = 24	Total regression: $P < 0.05$; Intercepts: $P = \text{ns}$; Slopes: $P < 0.05$
	Adjacent to crack	DI = 124.1 – 36.3(ln θ), s.e.(b) = 4.7, r = – 0.86***, n = 24	

^aDI, dispersion index (g kg^{–1}); θ , soil water content (g kg^{–1}); s.e.(b), standard error of the slope; r, correlation coefficient; n, number of observations; ** and ***, indicate significance at 0.01 and 0.001 probability levels, respectively; ns, nonsignificant.

TABLE 3 Effect of cracks on soil physical and chemical properties in Field C1 at Australian Cotton Research Institute, November 1997

Soil property	Location	0.0–0.3 m	0.3–0.6 m	0.6–0.9 m
GMD, mm ^a	Soil adjacent to cracks	5.3	7.0	6.4
	Bulk soil	6.1	6.9	7.6
	<i>P</i> <	ns ^b	ns	ns
pH (0.01M CaCl ₂)	Soil adjacent to cracks	7.3	7.3	7.3
	Bulk soil	7.2	7.2	7.2
	<i>P</i> <	ns	ns	ns
EC _{1:5} ^c , dS m ⁻¹	Soil adjacent to cracks	0.12	0.17	0.22
	Bulk soil	0.12	0.13	0.24
	<i>P</i> <	ns	ns	ns
Plastic limit, g kg ⁻¹	Soil adjacent to cracks	220	219	221
	Bulk soil	221	221	139
	<i>P</i> <	ns	ns	0.01
Soil organic C, g kg ⁻¹	Soil adjacent to cracks	8.0	6.5	5.9
	Bulk soil	8.4	6.5	5.0
	<i>P</i> <	ns	ns	0.05
CaCO ₃ equivalent ^d g kg ⁻¹	Soil adjacent to cracks	27	25	36
	Bulk soil	50	35	40
	<i>P</i> <	0.05	0.01	0.05
EC _{1:5} /Exch. Na ⁺	Soil adjacent to cracks	0.17	0.12	0.14
	Bulk soil	0.15	0.14	0.18
	<i>P</i> <	ns	ns	ns
CEC ^e , cmol _c kg ⁻¹	Soil adjacent to cracks	30.4	33.6	33.6
	Bulk soil	31.1	31.8	32.3
	<i>P</i> <	ns	ns	ns

^aGeometric mean diameter of aggregates formed after puddling and drying an air-dried sample.

^bNonsignificant.

^cElectrical conductivity of a 1:5 soil:water suspension.

^dCalcium and magnesium carbonates. Calcium carbonate accounts for > 90% of the carbonates in these soils.

^eCation exchange capacity.

Plastic limit was significantly correlated only with soil organic C content such that for all three sites the relationship as follows:

Plastic limit = 180.5 + 5.63SOC, s.e.(b) = 0.73, *r* = 0.62***, *n* = 96,

where plastic limit is in g kg⁻¹, SOC is soil organic carbon in g kg⁻¹, s.e.(b) is the standard error of the slope, *r* = correlation coefficient, and *n* = number of observations. The higher organic C in the soil adjacent to cracks at “Glenarvon” may be due in part to soil organic matter associated with soil aggregates and dissolved organic carbon being washed down into soil cracks during rainfall and irrigation (Skjemstad, Janik, and Taylor 1998). Crop residues may also fall into cracks. In addition crop roots become concentrated in the cracks (Dexter 1986a; Van Noordwijk et al. 1993). Observations made during October 1993 and 1995 indicated that high numbers of wheat roots were concentrated in soil cracks at “Glenarvon” (Hulugalle and Entwistle unpublished data). Nonetheless, the differences between SOC of cracked and bulk soils at “Glenarvon” are not as large as could be expected from the published literature (Skjemstad, Janik, and Taylor 1998; Van Noordwijk et al. 1993) and observations of root growth at this site. Similarly, differences in SOC between soil adjacent to cracks and bulk soil at the other two sites were either small

TABLE 4 Effect of cracks on soil physical and chemical properties in perennial pasture at Australian Cotton Research Institute, February 1999

Soil property	Location	0.0–0.3 m	0.3–0.6 m	0.6–0.9 m
GMD, mm ^a	Soil adjacent to cracks	2.3	5.3	6.2
	Bulk soil	3.3	4.8	5.8
	<i>P</i> <	ns ^b	ns	ns
pH (0.01M CaCl ₂)	Soil adjacent to cracks	6.6	6.9	7.2
	Bulk soil	6.6	7.1	7.2
	<i>P</i> <	ns	ns	ns
EC _{1:5} ^c , dS m ⁻¹	Soil adjacent to cracks	0.13	0.16	0.20
	Bulk soil	0.16	0.16	0.20
	<i>P</i> <	0.05	ns	ns
Plastic limit, g kg ⁻¹	Soil adjacent to cracks	254	280	283
	Bulk soil	289	268	278
	<i>P</i> <	ns	ns	ns
Soil organic C, g kg ⁻¹	Soil adjacent to cracks	17.9	12.1	8.8
	Bulk soil	19.8	12.6	9.1
	<i>P</i> <	ns	ns	ns
CaCO ₃ equivalent ^d g kg ⁻¹	Soil adjacent to cracks	34	21	54
	Bulk soil	37	44	36
	<i>P</i> <	ns	ns	ns
EC _{1:5} /Exch. Na ⁺	Soil adjacent to cracks	0.63	0.64	0.52
	Bulk soil	0.75	0.76	0.57
	<i>P</i> <	0.05	0.05	ns
CEC ^e , cmol _c kg ⁻¹	Soil adjacent to cracks	27.0	30.3	30.0
	Bulk soil	26.7	30.1	29.9
	<i>P</i> <	ns	ns	ns

^aGeometric mean diameter of aggregates formed after puddling and drying an air-dried sample.

^b Nonsignificant.

^c Electrical conductivity of a 1:5 soil:water suspension.

^d Calcium and magnesium carbonates. Calcium carbonate accounts for > 90% of the carbonates in these soils.

^e Cation exchange capacity.

(0.6–0.9 m depth in Field C1) or absent (Tables 3 and 4). This may be caused by a higher rate of crop residue and soil organic matter mineralization in the cracks at Field C1 and the pasture (Van Noordwijk et al. 1993).

Soil adjacent to cracks also had lower values of CaCO₃ at “Glenarvon” and Field C1 (Tables 1 and 3), and lower exchangeable Ca²⁺ at “Glenarvon” (see earlier discussion). These observations may be related to a higher rate of organic matter mineralization in the cracks. Microbial decomposition and associated aerobic respiration of soil organic matter in alkaline soils releases CO₂ as a by-product, which dissolves in water forming carbonic acid. The carbonic acid can then solubilize the native Ca-carbonates, releasing Ca²⁺, which may either become adsorbed onto the clay minerals or remain in soil solution (Chorom and Rengasamy 1997). We suggest that many of these ions are then leached through the soil cracks with rainfall or irrigation. This is in agreement with much of the published literature which suggests that leaching is greater through cracked than through bulk soil (Baer et al. 1995; Haria et al. 1994; McGarity, Hoult, and So 1984; Probert et al. 1987).

In summary, soil properties in soil adjacent to cracks at “Glenarvon” appear to have been affected by processes such as root growth and nutrient leaching. The

dominance of cracks as the major pathway of nutrient leaching and root growth is not entirely surprising in view of the tillage practices such as annual discing, chiseling, and ridging which took place at this site. Intensive tillage disrupts stable soil pores such as biopores and their continuity in clay soils (Hulugalle, Entwistle, and Lobry de Bruyn 1997; Jensen, Petersen, and Hansen 1997). In contrast, minimum or no-tillage maintains integrity of biopores (Jensen, Petersen, and Hansen 1997; Hulugalle, Entwistle, and Lobry de Bruyn 1997). Where biopores remain, they can function as alternative preferential flow pathways and facilitate root growth (Dexter 1986b; Hulugalle, Entwistle, and Lobry de Bruyn 1997; Jensen, Petersen, and Hansen 1997; Mitchell, Ellsworth, and Meek 1995). In soils with stable biopores, cracks may play less of a role in leaching and root growth (Mitchell, Ellsworth, and Meek 1995). This is probably why there were so few differences in soil properties between cracked and bulk soils in Field C1 (minimum tilled for approximately 12 years) and the perennial pasture (untilled for nearly 25 years) at ACRI.

Conclusions

With intensive tillage, exchangeable Ca^{2+} and CEC, and CaCO_3 , were lower, and organic C and plastic limit higher in soil adjacent to cracks than in soil away from cracks. More stable aggregates also occurred with intensive tillage in soil adjacent to cracks. These differences were fewer or absent when tillage intensity and frequency were either reduced (minimum tillage) or eliminated (untilled perennial pasture). We suggest that the similarity between soil adjacent to cracks and bulk soil under either minimum or no-tillage was due to the occurrence of stable, undisturbed biopores. These biopores can function as alternative pathways to soil cracks for root growth, preferential flow, and nutrient leaching. Consequently, in a soil where stable biopore networks exist, soil cracks play a less dominant role in leaching and root growth than in a soil where tillage disrupts biopores. However, when the biopore networks are disrupted and cracks become the most common pathway for preferential flow, associated nutrient leaching, and root growth, significant differences can occur between soil adjacent to cracks and bulk soil with respect to some physical and chemical properties.

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